

A SEQUENTIAL TEST FOR THE MEAN OF AN INVERSE GAUSSIAN DISTRIBUTION

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SUMMARY

It is shown that a sequential version of the fixed sample size likelihood ratio test of Seshadri & Shuster (1974) suffers no loss of power in an asymptotic sense while at the same time leading to an appreciable saving of observations when the null hypothesis is true.

1. Introduction

Let $X_1, X_2, \dots, X_n$ be i.i.d.r.v's with common three-parameter Inverse Gaussian distribution specified by the density function.

$$(1.1) \quad f(t; x, a, \sigma) = x(2\pi\sigma^2 t^3)^{-1/2} \exp[-(x - at)^2/2\sigma^2 t^3], \quad t > 0$$

where $x, \sigma > 0$ and $a \geq 0$ are constants. The likelihood ratio test (L.R.T.) of $H_0: a = 0$ vs. $H_1: a > 0$ has been considered by Nadas (1973) and by Seshadri & Shuster (1974). If the parameter $\tau = \sigma/x$ is known, the L.R.T. accepts H_0 if

$$(1.2) \quad \sum_{i=1}^n X_i \geq (n/\tau Z_{\alpha/2})^2$$

$$\text{where } \int_{Z_{\alpha/2}}^{\infty} (2\pi)^{-1/2} \exp(-u^2/2) \cdot du = \alpha/2$$

The motivation for the present paper comes from the simple fact that (1.2) holds if and

only if $\sum_{i=1}^k X_i \geq (n/\tau Z_{\alpha/2})^2$ for some $k = 1, 2, \dots, n$.

Hence, if the n observations are taken one at a time and H_0 is accepted at stage k ($\leq n$) if and only if $\sum_{i=1}^k X_i \geq (n/\tau Z_{\alpha/2})^2$ then this sequential procedure will

- (a) have the same power function as the fixed sample L.R.T. based on the observations $X_1, \dots, X_n$, and
- (b) in many cases require fewer than n observations to reach a decision.

Our purpose is to show, when $\tau = \sigma/x$ is unknown, that a sequential version of the corresponding fixed sample L.R.T. retains properties (a) and (b) above in an asymptotic sense. This is done in section 3. In section 2 we give some useful preliminary results and in section 4 we discuss the relative saving in observations afforded by the sequential procedure.

2. Preliminary results

Let $\{w(s), s \geq 0\}$ be a path-continuous Wiener process with drift 0 and variance 1, and define

$$m(x; a, \sigma) = \inf \{s \geq 0 \mid \sigma w(s) + a s = x\}, \quad x \geq 0.$$

Lemma 2.1

$$\begin{aligned} (i) \quad P \{m(x; a, \sigma) \leq t\} &= \Phi((at - x)/\sigma t^{1/2}) + \exp(2ax/\sigma^2) \Phi(-(at + x)/\sigma t^{1/2}), \quad t \geq 0 \\ &= F(t; x, a, \sigma), \text{ say,} \end{aligned}$$

where Φ denotes the standard normal c.d.f.

$$(ii) \quad dF(t; x, a, \sigma) = f(t; x, a, \sigma) \cdot dt \quad (\text{c.f. (1.1)}).$$

$$(iii) \quad E \{\exp(-\lambda m(x; a, \sigma))\} = \exp(x \sigma^{-2}(a - (a^2 + 2\sigma^2 \lambda)^{1/2})) \text{ for all } \lambda > 0.$$

Proof

C.F. N  das (1973).

Lemma 2.2

Let $[z]$ denote the integer part of z and let $y \geq 0$ be fixed. Then

$$P \{n^{-2} \sum_{i=1}^{[nz]} X_i \leq t_n \mid a = y n^{-1}\} \xrightarrow[n \rightarrow \infty]{} P \{m(xz; y, \sigma) \leq t\}$$

for any sequence $\{t_n\}$ with $t_n \rightarrow t > 0$.

Proof

Immediate from part (iii) of lemma 2.1

Define for $n \geq 2$:

$$S_n = \sum_{i=1}^n X_i, \bar{X}_n = n^{-1} S_n \text{ and } U_n = \tau^{-2} \sum_{i=1}^n (X_i^{-1} - \bar{X}_n^{-1}).$$

Recall that $\tau = \sigma/x$.

Lemma 2.3

For each $a \geq 0$ and $\tau > 0$:

- (i) U_n is chi-square distributed on $n - 1$ degrees of freedom.
- (ii) $P \{ |U_n - (n - 1)| > (n - 1)A \} \leq 2 \exp((n - 1)t(t/2 - A/\sqrt{2}))$ for all $n \geq 2$, $A > 0$ and $0 < t < 1/\sqrt{2}$.

Proof

- (i) is proved by Tweedie (1957).
- (ii) A straightforward calculation gives (using (i))

$$E \exp(t(U_n - (n - 1))/\sqrt{2}) \leq \exp((n - 1)t^2/2) \text{ for all } t < 1/\sqrt{2}.$$

By Chebyshev's inequality, then for $0 < t < 1/\sqrt{2}$:

$$P \{ U_n - (n - 1) > (n - 1)A \} \leq \exp((n - 1)t(t/2 - A/\sqrt{2})).$$

Similarly, for $0 < t < 1/\sqrt{2}$:

$$\begin{aligned} P \{ U_n - (n - 1) < -(n - 1)A \} &= P \{ (n - 1) - U_n > (n - 1)A \} \\ &\leq \exp((n - 1)t(t/2 - A/\sqrt{2})). \end{aligned}$$

Remark

Note once again the following simple fact which will be used repeatedly and without further mention in the sequel, viz.

$$P \{S_k \geq b \text{ for some } [nd] \leq k \leq [nz] \mid a = yn^{-1}\}$$

$$= P \{S_{[nz]} \geq b \mid a = yn^{-1}\}, \text{ valid for all } b, y > 0 \text{ and } 0 < d < z.$$

3. The procedures and their asymptotic power

If $\tau = \sigma/x$ is unknown, the fixed sample L.R.T. rejects H_0 if

$$\tau_n^2 S_n < (n/t_{n-1, \alpha/2})^2$$

where $t_{n-1, \alpha/2}$ is the upper 50 α per cent point of Student's t -distribution on $n - 1$ degrees of freedom and

$$\tau_n^2 = (n-1)^{-1} \sum_{i=1}^n (X_i^{-1} - \bar{X}_n^{-1}).$$

Define for $y \geq 0$ and $\Psi \geq 0$:

$$G(y; \Psi, \alpha) = 1 - \Phi(\Psi/Z_{\alpha/2} - yZ_{\alpha/2}) - \exp(2\Psi y)\Phi(-\Psi/Z_{\alpha/2} - yZ_{\alpha/2})$$

Theorem 3.1

$$(i) \quad P \{ \text{reject } H_0 \mid a = 0 \} = \alpha$$

$$(ii) \quad \lim_{n \rightarrow \infty} P \{ \text{reject } H_0 \mid a = \Psi\sigma^2/xn \} = 1 - G(1; \Psi, \alpha)$$

Proof

(i) is proved by Seshadri & Shuster (1974).

(ii) Let ϵ be arbitrary but fixed, $0 < \epsilon < 1/2$. Then

$$P \{ \text{reject } H_0 \mid a = \Psi\sigma^2/xn \}$$

$$\leq P \{ \tau_n^2 S_n < (n/t_{n-1, \alpha/2})^2 (1 - \epsilon)^{-1} \mid a = \Psi\sigma^2/xn \}$$

$$+ P \{ \tau_n^2 < \tau^2 (1 - \epsilon) \mid a = \Psi\sigma^2/xn \}$$

$$= I_1(n, \epsilon) + I_2(n, \epsilon).$$

By lemma 2.2

$$\lim_{\epsilon \rightarrow 0} \lim_{n \rightarrow \infty} I_1(n, \epsilon) = F((\tau Z_{\alpha/2})^{-2}; x, \Psi\sigma^2/x, \sigma) = 1 - G(1; \Psi, \alpha)$$

and by lemma 2.3, for $0 < t < \sqrt{2} \epsilon < 1/\sqrt{2}$,

$$\begin{aligned} I_2(n, \epsilon) &= P \{U_n - (n-1) < -(n-1)\epsilon \mid a = \Psi\sigma^2/xn\} \\ &\leq 2 \exp[(n-1)t(t/2 - \epsilon/\sqrt{2})] \\ &\xrightarrow[n \rightarrow \infty]{} 0. \end{aligned}$$

Hence $\lim_{n \rightarrow \infty} \sup P \{\text{reject } H_0 \mid a = \Psi\sigma^2/xn\} \leq 1 - G(1; \Psi, \alpha)$

Similarly, the inequality $P \{\text{reject } H_0 \mid a = \Psi\sigma^2/xn\}$

$$\begin{aligned} &\geq 1 - P \{\tau^2 S_n \geq (n/t_{n-1, \alpha/2})^2 (1 + \epsilon)^{-1} \mid a = \Psi\sigma^2/xn\} \\ &- P \{\tau_n^2 \geq \tau^2 (1 + \epsilon) \mid a = \Psi\sigma^2/xn\} \end{aligned}$$

yields $\liminf_{n \rightarrow \infty} P \{\text{reject } H_0 \mid a = \Psi\sigma^2/xn\} \geq 1 - G(1; \Psi, \alpha)$.

This completes the proof of the theorem.

Remark

Using the asymptotic normality of maximum likelihood estimates N  das (1973) has derived an approximation for the power of the test against a *fixed* alternative, c.f. his (21).

The sequential version of the above fixed sample L.R.T. stops with

$N(n) = \min \{N, n\}$ observations, where

$$N = \text{first } \kappa \geq 2 \text{ such that } \tau_\kappa^2 S_\kappa \geq (n/t_{n-1, \alpha/2})^2$$

H_0 is rejected if and only if $N > n$.

Theorem 3.2

For all $\Psi \geq 0$:

$$(i) \quad \lim_{n \rightarrow \infty} P \{\text{reject } H_0 \mid a = \Psi\sigma^2/xn\} = 1 - G(1; \Psi, \alpha)$$

$$(ii) \quad \lim_{n \rightarrow \infty} P \{n^{-1}N(n) \leq y \mid a = \Psi\sigma^2/xn\} = G(y; \Psi, \sigma) \text{ for } 0 \leq y < 1.$$

Proof

We shall show that

$$\lim_{n \rightarrow \infty} P \{N \leq [ny] \mid a = \Psi \sigma^2 / xn\} = G(y; \Psi, \alpha) \text{ for all } y \geq 0.$$

Theorem 3.2 follows easily from this. First, for fixed $0 < d < y$, define

$$K_1(n, d) = P \{ \tau_k^2 S_k \geq (n/t_{n-1, \alpha/2})^2 \text{ for some } [nd] \leq k \leq [ny] \mid a = \Psi \sigma^2 / xn \}$$

$$K_2(n, d) = P \{ \tau_k^2 S_k \geq (n/t_{n-1, \alpha/2})^2 \text{ for some } 2 \leq k \leq [nd] \mid a = \Psi \sigma^2 / xn \}$$

and note that

$$(3.1) \quad K_1(n, d) \leq P \{N \leq [ny] \mid a = \Psi \sigma^2 / xn\} \leq K_1(n, d) + K_2(n, d).$$

Now choose and fix ϵ , $0 < \epsilon < 1/2$. Then

$$\begin{aligned} K_1(n, d) &\leq P \{ \tau_{[ny]}^2 S_{[ny]} \geq (n/t_{n-1, \alpha/2})^2 (1 + \epsilon)^{-1} \mid a = \Psi \sigma^2 / xn \} \\ &\quad + P \{ \tau_k^2 \geq \tau^2 (1 + \epsilon) \text{ for some } [nd] \leq k \leq [ny] \mid a = \Psi \sigma^2 / xn \} \\ &= I_1(n, \epsilon) + I_2(n, \epsilon, d). \end{aligned}$$

As in the proof of theorem 3.1,

$$\lim_{\epsilon \rightarrow 0} \lim_{n \rightarrow \infty} I_1(n, \epsilon) \leq G(y; \Psi, \alpha)$$

and for fixed t , $0 < t < \sqrt{2} \epsilon < 1/\sqrt{2}$,

$$I_2(n, \epsilon, d) \leq 2 \sum_{k=[nd]}^{[ny]} \exp[(k-1)t(t/2 - \epsilon/\sqrt{2})] \xrightarrow{n} 0,$$

since the series $\sum_{k=2}^{\infty} \exp[(k-1)t(t/2 - \epsilon/\sqrt{2})]$ is convergent.

$$\text{Consequently} \quad \limsup_{n \rightarrow \infty} K_1(n, d) \leq G(y; \Psi, \alpha).$$

By an entirely analogous argument we may obtain $\liminf_{n \rightarrow \infty} K_1(n, d) \geq G(y; \Psi, \alpha)$

Hence:

$$(3.2) \quad \lim_{n \rightarrow \infty} K_1(n, d) = G(y; \Psi, \alpha).$$

Now choose and fix $A > 1$. Then

$$\begin{aligned} K_2(n, d) &\leq P \{ \tau_{[nd]}^2 \geq (n/t_{n-1, \alpha/2})^2 (1+A)^{-1} \mid a = \Psi \sigma^2 / x n \} \\ &\quad + P \{ \tau_k^2 \geq \tau^2 (1+A) \text{ for some } 2 \leq k \leq [nd] \mid a = \Psi \sigma^2 / x n \} \\ &= J_1(n, d, A) + J_2(n, d, A). \end{aligned}$$

For fixed t , $0 < t < 1/\sqrt{2}$, by lemma 2.3,

$$\begin{aligned} J_2(n, d, A) &\leq 2 \sum_{k=2}^{[nd]} \exp[(k-1)t(t/2 - A/\sqrt{2})] \\ &\leq 2 \sum_{k=2}^{[nd]} \exp[-(k-1)t(A-1)/\sqrt{2}], \text{ since } t < \sqrt{2}. \\ &\xrightarrow[n \rightarrow \infty]{} 2 \exp(-t(A-1)/\sqrt{2}) \cdot \{1 - \exp(-t(A-1)/\sqrt{2})\}^{-1} \\ &\xrightarrow[A \rightarrow 0]{} 0 \end{aligned}$$

and by lemma 2.2

$$J_1(n, d, A) \xrightarrow[n \rightarrow \infty]{} P \{ m(xd; \Psi \sigma^2 / x, \sigma^2) \geq (\tau_{\alpha/2}^2)^{-2} (1+A)^{-1} \} \xrightarrow[d \rightarrow 0]{} 0,$$

since $m(xd; \Psi \sigma^2 / x, \sigma^2) \xrightarrow[d \rightarrow 0]{d} 0$ in probability.

Therefore, letting $n \rightarrow \infty$, $d \rightarrow 0$ and then $A \rightarrow \infty$, we find

$$(3.3) \quad \lim_{d \rightarrow 0} \lim_{n \rightarrow \infty} K_2(n, d) = 0$$

The required result now follows from (3.1) by letting $n \rightarrow \infty$, then $d \rightarrow 0$, and using (3.2)-(3.3).

Remarks

1. Theorems 3.1 and 3.2(i) give the local asymptotic power of the two procedures and show that the sequential procedure does indeed retain properties (a) and (b) mentioned in section 1.
2. Theorem 3.2(ii) shows that $n^{-1}N(n)$ converges in distribution to a non-negative random variable Y with c.d.f.

$$P\{Y \leq y\} = \begin{cases} 0 & \text{for } y < 0 \\ G(y; \Psi, \alpha) & \text{for } 0 < y < 1 \\ 1 & \text{for } y \geq 1 \end{cases}$$

which is of mixed type with an atom $1 - G(1; \Psi, \alpha)$ at $y = 1$.

4. Comparison of sample sizes

In view of the preceding results, it makes sense to define the local relative efficiency (in the Pitman-sense) of the fixed sample procedure w.r.t. the sequential version as

$$e(\Psi, \alpha) = \lim_{n \rightarrow \infty} n^{-1} E\{N(n) | a = \Psi \sigma^2 / x_n\}$$

(The existence of the limit is an easy consequence of theorem 3.2(ii) and the bounded convergence theorem.) Since the sequential procedure provides for early acceptance of H_0 , one would expect the greatest relative saving in observations to be realized under H_0 . Indeed, some calculation gives

$$\begin{aligned} e(\Psi, \alpha) &= (2/\pi)^{1/2} (1 - \exp(-Z^2_{\alpha/2}/2))/Z_{\alpha/2} & \text{for } \Psi = 0 \\ &= 1 - G(1, \Psi, \alpha) & \text{for } \Psi > 0 \end{aligned}$$

and we see that for $\Psi > 0$

$$e(\Psi, \alpha) \rightarrow 1 \text{ as } \alpha \rightarrow 0, \text{ while}$$

$$e(0, \alpha) \sim (\pi \log \alpha^{-1})^{-1/2} \rightarrow 0 \text{ as } \alpha \rightarrow 0.$$

This last relation clearly demonstrates the asymptotic superiority of the sequential over

the fixed sample procedure under H_0 .

5. Monte Carlo results

A small Monte Carlo experiment was performed in order to check the speed of approximation in theorem 3.2(i) for the case $a = 0$. Since the joint distribution of

$$(\tau_k^2 S_k, k = 1, \dots, n)$$

does not depend on τ , we take $\tau = 1$ throughout. For $\alpha = 0.10, 0.05, 0.01$ and for $n = 5, 10$ the quantities $P \{\text{reject } H_0 \mid a = 0\}$ and $E \{N(n) \mid a = 0\}$ were estimated on the basis of 5 000 independent repetitions of the test procedure. The results are displayed in table 1 below, where $\hat{\alpha}$ and $\hat{E}$ denote the Monte Carlo estimates of the error probability and the expected sample size respectively.

TABLE 1

Estimates of error probabilities and expected sample sizes for $a = 0$

		$\hat{\alpha}$	$\hat{E}$
n = 5	$\alpha = 0.10$	0.0922	3.02
	0.05	0.0492	2.81
	0.01	0.0076	2.44
n = 10	$\alpha = 0.10$	0.0960	5.07
	0.05	0.0500	4.36
	0.01	0.0140	3.74

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